Zhiqiang Zang

Education

2018–2024 **Ph.D.**, *The University of Texas at Austin (UT Austin)*, Austin, TX, USA
Software Engineering and Systems — Advisor: Milos Gligoric

2014–2018 **B.E.**, *Beijing University of Posts and Telecommunications (BUPT)*, Beijing, China
Telecommunication Engineering

Industry Experience

Feb 2024–Present **Research Scientist**, *Meta Platforms*

Oct 2026–Present Ads Training Infra, Vancouver, BC, Canada
Sparse optimization.

May 2026–Oct 2026 AI Trainers Platform

Dec 2025–May 2026 MSL (Meta Superintelligence Labs)
LLM profiling, performance observability and debugging.

Jun 2025–Dec 2025 AI Observability Platform
GPU performance debugging, LLM training profiling, AMD training support.

Feb 2024–Jun 2025 AI Infra Efficiency, Menlo Park, CA, USA
Peformance and efficiency tooling, Holistic Trace Analysis (HTA).

May–Aug 2023 **Software Engineer Intern**, *Uber Technologies*, San Francisco, CA, USA
Supported contracts in a Go static analyzer called NilAway (`https://github.com/uber-go/nilaway`) to reduce false positive errors.

May–Aug 2022 **Software Engineer Intern**, *Meta Platforms*, Bellevue, WA, USA
Migrated a large-scale data analytic engine onto a new environment to improve performance and maintainability.

May–Aug 2020 **Research Intern**, *Fujitsu Laboratories of America*, ~~Sunnyvale, CA, USA~~ Remote
Improved software program repair examples by minimizing buggy/fixed pairs of programs.

Jul–Aug 2019 **Software Engineer Intern**, *NIO*, Beijing, China
Developed and improved a physical and visual simulator for autonomous vehicles (Unreal Engine).

Patents

[1] Hiroaki Yoshida, **Zhiqiang Zang**, Mukul R. Prasad. Generation of Software Program Repair Examples. US11099817B1. Aug 24, 2021.

Publications

[7] **Zhiqiang Zang**, Fu-Yao Yu, Aditya Thimmaiah, August Shi, and Milos Gligoric. Java JIT Testing with Template Extraction. In *International Conference on the Foundations of Software Engineering*, pages 51:1–51:23, 2024.

[6] **Zhiqiang Zang**, Aditya Thimmaiah, and Milos Gligoric. JOG: Java JIT Peephole Optimizations and Tests from Patterns. In *International Conference on Software Engineering, Tool Demonstrations Track*, pages 11–15, 2024.

+1 (737) 333 2408 • zang_zq@outlook.com • zzqpro.net • cptgit

[5] **Zhiqiang Zang**, Aditya Thimmaiah, and Milos Gligoric. Pattern-Based Peephole Optimizations with Java JIT Tests. In *International Symposium on Software Testing and Analysis*, pages 64–75, 2023.

[4] **Zhiqiang Zang**, Fu-Yao Yu, Nathaniel Wiatrek, Milos Gligoric, and August Shi. JAttack: Java JIT Testing using Template Programs. In *International Conference on Software Engineering, Tool Demonstrations Track*, pages 6–10, 2023.

[3] **Zhiqiang Zang**, Nathaniel Wiatrek, Milos Gligoric, and August Shi. Compiler Testing using Template Java Programs. In *International Conference on Automated Software Engineering*, pages 23:1–23:13, 2022. **ACM SIGSOFT Distinguished Paper Award**.

[2] Pengyu Nie, Marinela Parovic, **Zhiqiang Zang**, Sarfraz Khurshid, Aleksandar Milicevic, and Milos Gligoric. Unifying Execution of Imperative Generators and Declarative Specifications. In *International Conference on Object-Oriented Programming, Systems, Languages, and Applications*, pages 217:1–217:26, 2020.

[1] Ben Buhse, Thomas Wei, **Zhiqiang Zang**, Aleksandar Milicevic, and Milos Gligoric. VeDebug: Regression Debugging Tool for Java. In *International Conference on Software Engineering, Tool Demonstrations Track*, pages 15–18, 2019.

Selected Projects

JAttack `https://github.com/EngineeringSoftware/jattack`
A template-based compiler testing framework that 1) accepts as input a template Java program written in a domain-specific language in Java; 2) executes the template to generate concrete programs and 3) tests Java JIT compilers using the generated programs. JavaParser, OW2 ASM, DSL, JIT, Java. Team Project. Discovered CVEs: CVE-2020-14792, CVE-2022-21305, CVE-2023-22044, CVE-2023-22045.

JOG `https://github.com/EngineeringSoftware/jog`
A framework that facilitates developing Java JIT peephole optimizations: 1) accepts as input a pattern written in the same way that tests are written; 2) translates the pattern into C/C++ code as a JIT optimization pass, and generates Java tests for the optimization; 3) detects shadow relation between a pair of optimizations. JavaParser, Z3, DSL, JIT, Java. Team Project. 7 pull requests have been integrated into the master branch of OpenJDK.

VeDebug `https://github.com/EngineeringSoftware/VeDebug`
A Java debugging tool that 1) automatically sets breakpoints where the current execution diverges from the previously captured one, and 2) provides video player features e.g., speed up/slow down the replay. Bytecode instrumentation(OW2 ASM, javaagent), Java. Team project.

Honors & Awards

2022 ACM SIGSOFT Distinguished Paper Award [3], ASE 2022

2017 Outstanding Undergraduate Award Finalist, BUPT

2015–2016 Qualcomm Innovation Scholarship, BUPT

2014–2015 Qualcomm Innovation Scholarship, BUPT

Presentations

Jul 2023 *Pattern-Based Peephole Optimizations with Java JIT Tests* [5] at ISSTA 2023, Seattle, WA, USA

Oct 2022 *Compiler Testing using Template Java Programs* [3] at ASE 2022, Oakland Center, MI, USA

Teaching Experience

Fall 2023 Teaching Assistant, UT Austin, ECE 382V: *Programming Paradigms* (26 students)

Sep 2022 Guest lecture, *Compiler Testing using Template Java Programs* [3], UT Austin, ECE 382V: *Software Tests in the Era of Nondeterminism*

Fall 2021 Teaching Assistant, UT Austin, EE 379K: *Programming Paradigms* (38 students)

Spring 2021 Teaching Assistant, UT Austin, EE 360T: *Software Testing* (101 students)

Mar 2021 Guest lecture, *(JIT) Compiler Testing*, UT Austin, EE 360T: *Software Testing*

Spring 2020 Teaching Assistant, UT Austin, EE 312H: *Software Design and Implementation I* (35 students)

Fall 2019 Teaching Assistant, UT Austin, EE 312: *Software Design and Implementation I* (60 students)

Spring 2019 Teaching Assistant, UT Austin, EE 422C: *Software Design and Implementation II* (172 students)

Professional Service

CGO 2024 Artifact evaluation committee member, *International Symposium on Code Generation and Optimization*

ICSE 2023 External reviewer, *International Conference on Software Engineering*

Fall 2022 Co-organizer, *Joint UT-Cornell Software Engineering Seminar*

ICSE 2022 External reviewer, *International Conference on Software Engineering*

ASE 2021 External reviewer, *International Conference on Automated Software Engineering*

ISSTA 2020 Artifact evaluation committee member, *International Symposium on Software Testing and Analysis*

ISSTA 2020 External reviewer, *International Symposium on Software Testing and Analysis*

Technical Skills

- Programming Languages: Java, Bash, Python (everyday), C, Go, C++
- Tools: Emacs (everyday), Git/Mercurial (everyday), OW2 ASM, JavaParser, Holistic Trace Analysis (HTA), PyTorch